

Consider the following schemas: Sailors (sid, sname, rating, age), Reserves (sid, bid, day) Boats (bid, bname, color)
Write the following queries in relational algebra, tuple relational Calculus and domain relational calculus:

a) Find the name of sailors who have reserved boat 103.

- $\pi_{\text{sname}}(\sigma_{\text{bid}=103}(\text{Reserves} \bowtie \text{Sailors}))$
- $\{S.\text{sname} \mid S \in \text{Sailors}, R \in \text{Reserves}, S.\text{sid}=R.\text{sid}, R.\text{bid}=103\}$
- $\{\text{sname} \mid \exists \text{sid}, \text{age}, \text{rating}, \text{day} (\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age}) \wedge \text{Reserves}(\text{sid}, 103, \text{day}))\}$

b) Find the names and ages of sailors with a rating above 7.

- $\pi_{\text{sname}, \text{age}}(\sigma_{\text{rating}>7}(\text{Sailors}))$
- $\{(S.\text{sname}, S.\text{age}) \mid S \in \text{Sailors}, S.\text{rating}>7\}$
- $\{(\text{sname}, \text{age}) \mid \exists \text{sid}, \text{rating} (\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age}) \wedge \text{rating}>7)\}$

c) Find the sids of all sailors who have reserved red boats

- $\pi_{\text{sid}}(\text{Reserves} \bowtie (\sigma_{\text{color}='red'}(\text{Boats})))$
- $\{R.\text{sid} \mid R \in \text{Reserves}, B \in \text{Boats}, R.\text{bid}=B.\text{bid}, B.\text{color}='red'\}$
- $\{\text{sid} \mid \exists \text{bid}, \text{day}, \text{bname} (\text{Reserves}(\text{sid}, \text{bid}, \text{day}) \wedge \text{Boats}(\text{bid}, \text{bname}, 'red'))\}$

d) Find all sids of sailors who have a rating of 10 or reserved boat 104.

- $\pi_{\text{sid}}(\sigma_{\text{rating}=10}(\text{Sailors})) \cup \pi_{\text{sid}}(\sigma_{\text{bid}=104}(\text{Reserves}))$
- $\{S.\text{sid} \mid S \in \text{Sailors}, S.\text{rating}=10\} \cup \{R.\text{sid} \mid R \in \text{Reserves}, R.\text{bid}=104\}$
- $\{\text{sid} \mid (\exists \text{sname}, \text{rating}, \text{age} (\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age}) \wedge \text{rating}=10)) \vee (\exists \text{bid}, \text{day} (\text{Reserves}(\text{sid}, 104, \text{day})))\}$

e) Find the names of sailors who have reserved boat 110.

- $\pi_{\text{sname}}(\sigma_{\text{bid}=110}(\text{Reserves} \bowtie \text{Sailors}))$
- $\{S.\text{sname} \mid S \in \text{Sailors}, R \in \text{Reserves}, S.\text{sid}=R.\text{sid}, R.\text{bid}=110\}$
- $\{\text{sname} \mid \exists \text{sid}, \text{rating}, \text{age}, \text{day} (\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age}) \wedge \text{Reserves}(\text{sid}, 110, \text{day}))\}$

f) Find the names of sailors who have reserved a red boat.

- $\pi_{\text{sname}}((\text{Reserves} \bowtie (\sigma_{\text{color}='red'}(\text{Boats}))) \bowtie \text{Sailors})$
- $\{S.\text{sname} \mid S \in \text{Sailors}, R \in \text{Reserves}, B \in \text{Boats}, S.\text{sid}=R.\text{sid}, R.\text{bid}=B.\text{bid}, B.\text{color}='red'\}$
- $\{\text{sname} \mid \exists \text{sid}, \text{rating}, \text{age}, \text{bid}, \text{bname}, \text{day} (\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age}) \wedge \text{Reserves}(\text{sid}, \text{bid}, \text{day}) \wedge \text{Boats}(\text{bid}, \text{bname}, 'red'))\}$

g) Find the sname, bid, and day for each reservation.

- $\pi_{\text{sname}, \text{bid}, \text{day}}(\text{Reserves} \bowtie \text{Sailors})$
- $\{(S.\text{sname}, R.\text{bid}, R.\text{day}) \mid S \in \text{Sailors}, R \in \text{Reserves}, S.\text{sid}=R.\text{sid}\}$
- $\{(\text{sname}, \text{bid}, \text{day}) \mid \exists \text{sid}, \text{rating}, \text{age} (\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age}) \wedge \text{Reserves}(\text{sid}, \text{bid}, \text{day}))\}$

Consider the following schema: Supplier (sid, sname, city) Part (pid, color, weight) Supply (sid, pid, quantity)
Compute the following queries in relational algebra, tuple relational Calculus and domain relational calculus:

a) Find the sids of supplier who supplies parts p1 and p2.

- $\pi_{\text{sid}}(\sigma_{\text{pid}='p1'}(\text{Supply})) \cap \pi_{\text{sid}}(\sigma_{\text{pid}='p2'}(\text{Supply}))$
- $\{S.\text{sid} \mid S \in \text{Supplier}, \exists Q1, Q2 (\text{Supply}(S.\text{sid}, 'p1', Q1) \wedge \text{Supply}(S.\text{sid}, 'p2', Q2))\}$
- $\{\text{sid} \mid \exists Q1, Q2 (\text{Supply}(\text{sid}, 'p1', Q1) \wedge \text{Supply}(\text{sid}, 'p2', Q2))\}$

b) Find the colors of part screw and bolt.

- $\pi_{\text{color}}(\sigma_{\text{pid}='screw' \vee \text{pid}='bolt'}(\text{Part}))$
- $\{P.\text{color} \mid P \in \text{Part}, (P.\text{pid}='screw' \vee P.\text{pid}='bolt')\}$
- $\{\text{color} \mid \exists \text{pid}, \text{weight} (\text{Part}(\text{pid}, \text{color}, \text{weight}) \wedge (\text{pid}='screw' \vee \text{pid}='bolt'))\}$